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Flow detection circuit

Abstract

A dispensing system for a beverage comprises in a tap system a bore for housing a duct. Along the bore, close to or on the duct, at least two electrodes are provided such that at least at some locations along the duct, the two electrodes are provided opposite to one another with the duct in between, thus constituting a capacitor. An oscillating signal is provided to one electrode and a signal is read out from the other electrode. As a beverage is drawn through the duct in a container, capacitance of the capacitor changes. The flowing beverage may have different characteristics, but capacitance may also change as the beverage in the duct is in conducting contact with a container that may be in contact with an earth contact. The change of capacitance results in a change of the amplitude of a detection circuit connected to the second electrode.

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Background/Summary

TECHNICAL FIELD

(1) The various aspects and embodiments thereof relate to a circuit and device for detection of a flow through a duct as part of a system for dispensing beverages, using a capacitive sensor.

BACKGROUND

(2) U.S. Pat. No. 6,545,488B2 discloses a system for capacitive flow measurement through a duct in a large transportation system. Two electrodes are helically wound around the duct and a signal is applied to the electrodes. Upon change of flow through the duct, a frequency of a signal in a detection circuit varies. The detection is transformed to a voltage signal, which is used for further processing.

SUMMARY

(3) When using variation of capacitance for changing an oscillating frequency, use of an inductance is highly preferred. Inductances are bulky and relatively expensive circuit components.

Furthermore, for further conversion of frequency changes to a voltage signal, a significant amount of circuitry is required. It is preferred to provide a more efficient arrangement for detecting flow through the duct of a beverage dispensing system.

(4) A first aspect provides a circuit for detecting a flow through a duct of a tap for dispensing beverages. The circuit comprises an alternating signal source having a signal terminal arranged to provide an alternating signal to a capacitive element comprising two electrodes provided at opposite sides of the duct along at least part of the length of the duct and a detection circuit. The detection circuit is arranged to be connected to the capacitive element and arranged to detect a signal amplitude value, the detection circuit being arranged to provide a detection signal based on the signal amplitude value at a detection terminal comprised by the detection circuit. The flow detection circuit also comprises a processing circuit arranged to receive the detection signal, determine whether the detection signal satisfies a pre-determined criterion; and provide a flow signal if the pre-determined criterion is met.

- (5) This system works with one frequency throughout the circuit, thus reducing complexity. Furthermore, it may be implemented in the analogue domain, in the digital domain or partially in the analogue domain and partially in the digital domain without departing from the scope of this aspect.
- (6) In an implementation, the alternating signal source has a signal terminal arranged to be connected to a first capacitor electrode provided along at least part of a length of the duct and the detection circuit is arranged to be connected to a second capacitor electrode provided along at least part of a length of the duct for receiving an electrode signal.
- (7) A further implementation further comprises an adding circuit for summing the detection signal and a reference control signal for providing a controlled detection signal, wherein the processing circuit is further arranged to provide a reference control signal based on the controlled detection signal.
- (8) The skilled person will understand that a subtractor also falls within the definition of an adder. With this implementation, large variation in the amplitude of the signal at the second electrode may be controlled within boundaries such that detection of flow and characteristics of the flow may be more convenient.
- (9) In another implementation, the detection circuit comprises a peak detection circuit and the detection signal is based on the output of the peak detection signal. Whereas other types of circuit may be used for determining an amplitude value, like multipliers, a peak detector is preferred for determining amplitude.
- (10) In yet another embodiment, the detection circuit is arranged to detect fluctuations of charge on the second electrode and to provide a voltage signal based on the detected fluctuations. Whereas the circuit may be implemented as current based, voltage based circuitry is preferred.
- (11) A second aspect provides a casing for housing a duct for a beverage dispensing tap. The casing comprises an elongate bore having a proximal opening and a distal opening for housing and guiding the duct, a first electrode provided over at least part of the length of the bore and a second electrode provided over at least part of the length of the bore. In this casing, the first electrode is arranged to be connected to the signal terminal of a circuit according to any of the preceding claims; and the second electrode is arranged to be connected to the detection terminal of a circuit according to the first aspect. Such casing comprises the electrodes preferred for operation of the circuit according to the first aspect, providing an advantageous arrangement for housing a duct for dispensing beer—or another beverage—and detection flow of the beer.
- (12) In an embodiment, the casing comprises a first shell and a second shell, the first shell comprising a first elongate recess and the second shell comprising a second elongate recess such that when the first shell and the second shell are joined for forming the case, the first recess and the second recess provide at least part of the duct. This casing is advantageous for use of a disposable duct, which may be conveniently inserted and removed from the bore. Furthermore, if the casing is provided by means of two halves, electrodes and circuitry may all be provided on one and the same shell part.
- (13) A third aspect provides a dispensing device for dispensing a beverage. The device comprises the casing according to the second aspect, the circuit according to the first aspect. In this device, the first electrode is connected to the signal terminal; and the second electrode is connected to the detection terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The various aspects and implementations thereof will now be elucidated in conjunction with drawings. In the drawings:

- (2) FIG. 1: shows a beverage dispensing system; and
- (3) FIG. 2A: shows a first flow detection circuit;
- (4) FIG. 2B: shows a second flow detection circuit;
- (5) FIG. 3: shows another flow detection circuit; and
- (6) FIG. 4: shows an electrical circuit equivalent of the other flow detection circuit.

DETAILED DESCRIPTION

(7) FIG. 1 shows a beer dispensing system **100** as an implementation of a dispensing device for dispensing beverages. The beer dispensing system **100** comprises a duct **120** for providing or forming a channel for beer to flow from a reservoir like a keg or a large tank (not shown) to a dispense valve **122**. The valve **122** can be an integral part of the duct, or the **120** duct can be coupled to the valve **122**. The duct **120** is or at least comprises preferably a flexible and more preferably a resilient tube comprising an organic polymer, like PVC, silicon, polyethylene, other, or a combination thereof.

(8) The duct **120** is provided in and extends through a bore **114** in a casing **112** that constitutes a stand for a dispense tap **116**. The dispense tap **116** comprises a tap handle **118** for operating the dispense valve **122**. Within the casing **112**, a first electrode **132** and a second electrode **134** are provided. The first electrode **132** and the second electrode **134** are provided such that they are located at opposite sides of the duct **120** for at least a part of the length of the duct **120**.

(9) Such constellation may be achieved by providing the first electrode **132** and the second electrode **134** in the casing, at opposite sides at the inner wall of the bore **114** and substantially parallel to the length of the bore **114** and the duct **112**. In this constellation, the electrodes are provided opposite to one another over their full lengths. Alternatively, the electrodes are provided in any other way such that they are provided at opposite sides at least for one or more parts of the length of the bore **114**.

(10) The beer dispensing system further comprises a flow detection circuit **200** for detecting whether beer is drawn from the reservoir using the beer dispensing system **100**. The flow detection circuit **200** is connected to the first electrode **132** and the second electrode **134**.

(11) The casing may be provided by means of two shells to facilitate removal and insertion of the duct **120**. This is particularly advantageous if the duct **120** is a disposable duct. In such embodiment, the first electrode **132** and the second electrode **134** are preferably provided on the same shell part, together with a flow detection circuit **200**. The duct **120** may be embedded in a first shell having a recess as part of the bore **114**.

(12) Subsequently, a second shell is joined with the first shell for forming the casing **112**.

Alternatively, the duct **120** is led through the bore **114** with a closed bore, i.e. with the two halves joined, or, if the casing **112** mainly comprises a single unit through which the bore is provided. The duct **120** may be inserted in the bore **114** from above or below, in the constellation shows by FIG.

1. The bore **114** may be provided inside a massive casing **112** or as a tube or pipe, either rigid or flexible, in a hollow casing **112**. The shape and dimension of the cross-section of the bore **114** may vary or be substantially the same over the length of the bore **114**.

(13) The duct **120** is preferably provided inside the bore **114** such that it does not contacts the first electrode **132** and the second electrode **134**. Whereas the duct **120** is preferably provided as a flexible dispensing line predominantly comprising an organic polymer that is electrically insulating, liquid and water condensation in particular, may provide a conductive path. Therefore, the duct **120** is provided at a distance from the first electrode **132** and the second electrode **134**.

Alternatively or additionally, the first electrode **132** and the second electrode **134** are provided with an insulating film, at least at a side facing the inner space of the bore **114**.

(14) The bore **114** may be provided as a rigid guide for the duct **120** or as a flexible guide. In the latter case, the bore **114** may be provided comprising a flexible tubing inside which tubing the duct **120** may be provided.

(15) FIG. 2A shows the flow detection circuit **200** in further detail. FIG. 2A shows functional

components preferred for implementing the flow detection circuit **200**. The flow detection circuit **200** comprises a signal generator **202** for providing an alternating signal, as a voltage source or a current source. Preferably, the signal generator **202** generates a sine wave, preferably at a frequency between 1.5 kHz and 3 kHz, more preferably between 2 kHz and 2.5 kHz and most preferably at 2.3 kHz. It is noted that depending on the values of the various components, other values of the frequency may be selected, between 1 kHz and 4 kHz, between 5 kHz and 10 kHz, between 10 kHz and 50 kHz, between 50 kHz and 100 kHz, between 100 kHz and 102 kHz, or above, even up to 102 kHz and up.

(16) In another implementation, the signal generator generates another wave form, including at least one of a triangular signal, a block wave, a saw tooth signal, other, or a combination thereof. The frequency of the signal is preferably set, though it may be variable.

(17) The generated signal is applied to the first electrode **132**. The signal may optionally be provided via a buffer, decoupling capacitor, a resistor, another type of impedance or a combination thereof. Together with the second electrode **134**, the first electrode **132** constitutes a capacitor **204**. Variations of charge on the first electrode **132** due to the alternating signal applied will result in variations of charge on the second electrode **134**—as is the basic principle of a capacitance like the capacitor **204**.

(18) The fluctuations in charge on the second electrode **134** result in an alternating current at the second electrode **134**. The amount of the fluctuation depends on characteristics of the medium between the first electrode **132** and the second electrode **134**. Such characteristics include, but are not limited to, distance between the electrode, permittivity of the medium or media between the electrodes and other characteristics of the medium or media.

(19) The alternating current is provided to an amplifier and preferably to a trans impedance amplifier **206** for converting the alternating current to an alternating voltage. The alternating voltage has the same frequency as the signal provided by the signal generator **202**. The amplified voltage signal is provided to a band stop filter or notch filter **208** which has a centre frequency substantially equal to a frequency of mains voltage supply. For the majority of the world this is 50 Hz, for some regions including the Americas this is 60 Hz. At some locations, other frequencies may apply. Alternatively or additionally, for reducing influence of the mains power grid, the casing **112** may be provided with shielding for reducing electromagnetic interference on the flow detection circuitry.

(20) The alternating signal is subsequently provided to a bandpass filter **210**. The bandpass filter **210** has a centre frequency substantially equal to the frequency of the signal provided by the signal generator **202**. The bandpass filter **210** may be provided comprising capacitances and inductances as to define slopes and centre frequency independently. However, as inductances may be bulky and relatively expensive, use of a multiple feedback bandpass filter (MFB filter) is preferred.

(21) The signal thus filtered is provided to a rectifier **212**. The rectifier **212** is in this implementation a half-bridge rectifier. Alternatively, the rectifier **212** may be implemented as a full bridge rectifier. However, the input signal is a data signal of which only the top output level of the rectified signal is relevant, for which reason not the full signal is required to be rectified—as opposed to a power signal.

(22) The rectified signal is provided to a level detection circuit **214** for detecting a peak level of the rectified signal. Of the filtered and rectified signal, the peak level is determined and the level detection circuit and the output of the peak detector **214** is provided to an amplifier **216**.

(23) The amplified signal at the output of the amplifier is provided to a feedback control loop. The feedback control loop comprises an adder **218**—which may with minor design modifications also be implemented as a subtractor—and a control circuit **220**. The control circuit **220** is preferably implemented as a PID control circuit **220**—a proportional—integral—derivative controller.

(24) The PID control circuit **220** is preferably provided as part of a microcontroller **240** as a processing module. The processing module may also be embodied as a microprocessor or central processing unit or any other electronic computation circuit arranged for executing computer

readable instructions and suitable to manufacture any device according to the various aspects and embodiments thereof. Hence, the output of the adder **218** is sampled and digitised prior to being provided to the PID control circuit **220**. The output of the PID control circuit **220** is initially digital and converted to the analogue domain prior to being provided to the adder **218**.

(25) A reason for providing the feedback control loop is that depending on the container in which beer is dispensed into, such as a glass or a jug, the variations in the signals may vary significantly in magnitude. The feed control loop ensures the analogue output of the circuit as provided to the microcontroller is provided in the proper range, for example between 0 and 5 Volt, between 0 and 3.3 Volt, between 0 and 2.5 Volt or another range.

(26) The digitised signal provided to the PID control circuit **220** is also provided to a central control circuit **242**. The central control circuit **242** is part of the microcontroller **240** of which the functionality may be programmed or already available upon manufacturing. The central control circuit evaluates the digitised output signal of the adder **218**—the controlled signal—to one or more pre-determined values. These values may be stored in a storage module **244**, either provided separately or as part of the microcontroller **240**.

(27) The values may also be adjusted based on ambient temperature. To that purpose, the microcontroller **240** is connected to a temperature sensor **250**. The temperature sensor **250** may be provided close to the duct **120**, for example near or between the first electrode **132** and the second electrode **134**. In that configuration or a configuration equivalent thereto, the temperature sensor **250** may be used to measure a temperature of the beverage being dispensed. This allows for control of quality of the beverage dispensed—with beer, the serving temperature is very important for the total experience of drinking a premium lager beer—and it allows for monitoring of a cooling system for cooling the beverage.

(28) If the controlled signal is above or below a particular pre-determined value, it may be determined that beer is dispensed from the beer dispensing system **100**. It has been determined that as beer is being dispensed from the beer dispensing system **100**, the level of the alternating current at a terminal of the second electrode varies. This variation of the amplitude of the received capacitor current depends on a type of container used—glass, jug, pitcher or bucket—and how the container is being held—by the full hand or by the tips of a few fingers only.

(29) Hence, as the controlled signal is at a particular value, it may firstly be detected that beer is being drafted and in a container in particular, such as a pitcher, a jug or a glass. Second, it may be established in what type of container—pitcher, glass, jug—beer is being drafted. Third, it may be established how the container is being held. Fourth, as detection is based on the duct **120** being present in the bore **114**, between the first electrode **132** and the second electrode **134**, it may be detected whether the duct **120** is present at all.

(30) Based on the determinations, further data processing may be executed. A proper beer is being drafted by fully opening the valve **122** of the tap **116**, commonly executed by swivelling the tap handle **118** by e.g. about ninety degrees. This means that the dispense valve is either open or closed; the flow is either maximal or zero. If the period in time of drafting may be determined based on processing of the controlled signal (or another signal in the chain) is multiplied by the maximum flow, the amount of beer drafted may be calculated. In this way, a pre-warning may be issued in case a keg or other reservoir is almost empty.

(31) The microcontroller **240** also comprises an actuator circuit **246** which may be programmed or already available upon manufacturing. The actuator circuit is arranged to control an actuator outside the microcontroller **240**, like the light emitting diode **260**. The light emitting diode **260** may provide a lighting functionality for lighting the casing **112** or part thereof. Alternatively or additionally, other light sources, a display screen for showing text, video or still images, an audio source or other peripheral devices may be controlled.

(32) With the type of container determined, the type of container may be shown on a screen close to the dispensing system **100**. And if it is determined how the container is being held, feedback may

be provided to a person using the dispensing system **100**. For example, if it is preferred a glass is being held by the tips of fingers rather than the full hand, the user may be informed to take corrective action if it is determined the glass is being held by the full hand.

(33) Another action may be taken by a remotely located actuator. To that purpose, the flow detection circuit comprises a communication module **248** connected to the central control circuit **242**—or another part of the microcontroller **240**. Instructions to remotely located actuators may be provided to the communication module **248**, which transfers the instructions using a protocol like IEEE 802.11, popularly known as WiFi, Zigbee, Bluetooth, LoRa, an LTE protocol, other, or a combination thereof. Alternatively or additionally, data may be received via the communication module **248**, for example for programming the microcontroller **240**.

(34) The dispensing system **100** thus provided is preferably used for dispensing beer, but it may also be used for dispensing other beverages like cider, alcopops or soft drinks.

(35) The filters used in the flow detection circuit **200** are preferably active filters, which may be implemented using commercially available operational amplifiers.

(36) In the embodiments discussed above, the first electrode **132** and the second electrode **134** are provide at either side of the bore **114**—or the duct **112**, for that matter—and the first electrode **132** is connected to the signal generator **202** and the second electrode **134** is connected to the signal detection circuit right of the capacitive element the capacitor **204** embodies. In another embodiment, the capacitor **204** is provided in a capacitive divider, in series with or parallel to a further capacitive element having a substantially fixed capacity value.

(37) As the capacity value of the capacitor **204** changes upon beer being drawn through the duct **112** and amplitude of a signal provided by the signal generator changes, depending on the configuration, a change of current through or voltage over the capacitor **204** and/or through/over the further capacitor may change. This change may be detected and used to determine whether a liquid flowing through the duct is dispensed in a container. In case the capacitors are connected in parallel, a change in current is to be detected and in case the capacitors are connected in series, a change in voltage is to be detected. It is noted that this relates to the amplitude of the signal, rather than the actual value.

(38) FIG. 2 B shows an alternative embodiment of the flow detection circuit **200**. Part also present in the embodiment disclosed by FIG. 2A and variations thereof discussed above with the same reference sign as in FIG. 2 A have equivalent functionality and will not be discussed in further detail, unless required for better intelligibility.

(39) The flow detection circuit depicted by FIG. 2 B comprises an analogue to digital converter **272** and a digital to analogue converter, connected to a microcontroller **280** (equivalent to the microcontroller **240** of FIG. 2A)—which may also be implemented using a microprocessor or an equivalent electronic computational device.

(40) The microcontroller **280** comprises, either in a soft-programmed fashion or a hardwired fashion, a Fast Fourier Transform (FFT) module **284** for applying a discrete transformation of a signal to a frequency domain, a signal generator **282**, a convolution module **286** and a step function generator **288**. In another alternative, one or more of these parts may be implemented in a separate circuit. The other parts of the microcontroller **280** are the same as or similar or equivalent to parts of the microcontroller **240** depicted in and discussed in conjunction with FIG. 2A.

(41) In this embodiment, the digital signal generator **282** is a digital signal generator and implemented in the microcontroller **280**. The signal, as in FIG. 2A, is preferably provided at 7.5 kHz, but may vary between 5 kHz and 10.sup.1 kHz, between 10.sup.1 kHz at 5.Math.10.sup.1 kHz or above, even up to 102 kHz and up. Alternatively, the frequency may vary between 1.5 kHz and 3 kHz, more preferably between 2 kHz and 2.5 kHz and most preferably at 2.3 kHz.

(42) The generated signal preferably has a sine waveform, but may also have a square, triangle, saw tooth or other waveform or any combination thereof. The signal generated by the digital signal generator is transformed to the analogue domain by means of the digital to analogue converter **274**

and provided to the first electrode **132**.

(43) The signal received from the second electrode **134** and provided by the band stop filter **208** is converted to the digital domain by means of the analogue to digital converter **272** and provided to the microprocessor **280**. In the microprocessor **280**, the digitised signal is converted to the frequency domain by means of the FFT module **284**, using the signal generated by the digital signal generator **282**. This means that the FFT module **284** and the capacitor **204** are provided with one and the same signal, at the same frequency. This allows for improved detection of the signal provided to the first electrode **132** and received by the second electrode **134**—and changes in that signal due to liquid flowing through the duct **120** or not. And this allows the signal transferred from the digital signal generator **282**, via the capacitor **204**, to be filtered out from the data received from the analogue to digital converter **272**, as both signals have the same frequencies.

(44) The signal provided by the FFT module **284** is provided to the convolution module **286** and convoluted with a step function provided by the step function generator **288**. The step function generator **288** does not necessarily have to be an actual signal generator, but may also be implemented as values stored in a memory and as such provided to the convolution module **286**.

(45) The convoluted signal as well as the direct output of the FFT module **284** are provided to the central control circuit **242**. The convoluted signal is used for detection whether the liquid in the duct **120** flows or not. The direct output of the FFT module **284** is used to detect a level of the signal received from the second electrode **134**, as received from the analogue to digital converter **272**.

(46) FIG. **3** shows a sensing arrangement **300** as another implementation of a sensing setup according to the aspects discussed above for use in, for example, the beer dispensing system **100**. FIG. **3** shows the sensing arrangement **300** comprising beer conduit module **302** and a sensing module **304**. The beer conduit module **302** comprises a beer conduit **312** connected at a distal end to a beer keg **306** arranged to hold beer—or another dispensable liquid—and connected at a proximal end to a tap like the dispense tap **116** as shown by FIG. **1**.

(47) Around the beer conduit **312**, a transmitter electrode **314**, a first receiving electrode **316** and an optional second receiving electrode **318** are provided. Between the transmitter electrode **314** and the first receiving electrode **316**, an optional first shielding electrode **320** is provided and between the first receiving electrode **316** and the second receiving electrode **318**, an optional second shielding electrode **322** is provided. The electrodes are preferably not provided in direct contact with the beer conduit, but this may be the case in another implementation.

(48) The electrodes preferably fully surround the beer conduit **312**. In an implementation in which the casing **112** (FIG. **1**) is provided having two or more shell parts, the electrodes may be constituted as comprising multiple electrode parts, wherein each part is comprised by a shell part of the casing **112**. With the casing **112** assembled, the electrode parts of each electrode are in conductive contact with one another—but not with parts of other electrodes. In another implementation, not all shell parts comprise parts of each electrode, in which case each electrode may not fully surround the beer conduit **312**.

(49) Preferably, the shielding electrodes, if present, are—measured along the length of the beer conduit **312**—preferably shorter than the transmitter electrode **314**, the first receiving electrode **316** and the second receiving electrode **318**. The shielding electrodes are connected to ground level or to the zero reference of the signal source **V1**.

(50) The sensing module **304** comprises a signal source **V1**, a first reference capacitor **C1** and a second reference capacitor **C2**. The signal source **V1** is substantially the same, similar or at least equivalent to the signal generator **202** of FIG. **1**, unless indicated otherwise.

(51) The signal source **V1** is at a first terminal connected to earth or a zero reference and at a second terminal connected to the transmitter electrode **314** and a first terminal of the second reference capacitor **C2**. Between a second terminal of the second reference capacitor **C2** and a first terminal of a first reference capacitor **C1**, a reference voltage terminal is provided, with a reference

voltage. A second terminal of the first reference capacitor C1 is connected to ground level or the zero reference terminal of the signal source V1.

(52) The sensing module 304 comprises a first signal comparator 342 and a second signal comparator 344. The signal comparators are arranged to compare analogue input signals and to provide an output signal in response to the comparing operation. The signal comparators may be implemented using analogue operational amplifier circuits. Alternatively or additionally, the comparing of signals may be executed in a digital domain. To that end, the signal comparators may comprise analogue to digital converters. The first signal comparator 342 and a second signal comparator 344 may be implemented equivalent to the flow detection circuit 200 as depicted by and discussed in conjunction with FIG. 2A and FIG. 2 B. In such implementation, the transmitter electrode 314 and the first receiving electrode 316 form a first capacitor equivalent to the capacitor 204 as shown by FIG. 2A and FIG. 2 B; the transmitter electrode 314 and the second receiving electrode 318 form a second capacitor also equivalent to the capacitor 204 as shown by FIG. 2A and FIG. 2 B

(53) The comparing by the signal comparators may take place instantaneously, in a continuous time domain. Alternatively or additionally, the comparing may take place in an amplitude domain, comparing amplitudes of the reference voltage and a voltage as sensed by means of the first receiving electrode 316 and the second receiving electrode 318. Such comparing in the amplitude domain may takes place continuously or at regular intervals, for example every fifth, tenth or twentieth of a second to every one or two seconds.

(54) A first signal comparator 342 is connected to the first receiving electrode 316 and the reference voltage terminal. A second signal comparator 344 is connected to the second receiving electrode 318 and the reference voltage terminal. In this configuration, the first signal comparator 342 compares voltages on the first signal electrode to 316 the reference voltage and the second signal comparator 344 compares voltage on the second signal electrode 318 to the reference voltage.

(55) The comparators provide an output signal based on a difference between the input voltages. Further signal processing, prior to comparing, after comparing or both, may take place, in a way as discussed in conjunction with FIG. 2A, FIG. 2 B, otherwise, or a combination thereof.

(56) FIG. 3 shows a parasitic capacitance C6 from the beer conduit 302 to ground to model a connection between the beer conduit 302 and ground via a person drawing a beer. Furthermore, a keg parasitic capacitance C7 is drawn to model a capacitance from the beer conduit 302 to ground via the beer keg 306 and the beer conduit 302.

(57) FIG. 4 shows an equivalent circuit diagram 400 of the configuration depicted by FIG. 3. FIG. 4 shows the first reference capacitor C1 and the second reference capacitor C2. FIG. 4 also shows a first receiving terminal Rx.sub.1 providing an input to the first comparator 342 and a second receiving terminal Rx.sub.2 providing an input to the second comparator 344. The transmitting electrode 314 is modelled as a transmitting capacitance C3, the first receiving electrode 316 is modelled as a first receiving capacitance C4 and the second receiving electrode 318 is modelled as a second receiving capacitance C5.

(58) The beer conduit 312 is modelled as a chain of resistances; a first conduit resistance R1 between the transmitting electrode 314 and the first receiving electrode 316 and a second conduit resistance R2 between the first receiving electrode 316 and the second receiving electrode. The proximal end of the beer conduit 312 up to the dispense tap 116 is modelled as a third conduit resistance R3 and the dispense tap 116 itself is modelled as a tap resistance R5. The distal end of the beer conduit 312 between the transmitting electrode 314 and the keg 306 is modelled as a fourth conduit resistance R4 and the resistance over the keg to keg parasitic capacitance C7 is modelled as a keg resistance R6.

(59) During a dispensing operation of the beer dispensing system 100, while dispensing beer, at least one of the modelled conduit resistances and the transmitting electrode 314 and the receiving electrodes changes. As a result, the voltages at the first receiving terminal Rx.sub.1 and the second

receiving terminal Rx.sub.2 change. With the values of first reference capacitance C1 and the second reference capacitance C2 being substantially constant, the reference voltage is substantially constant.

(60) Variation of at least one value of capacitances and/or resistances at one side of the bridge and no substantial change at the other side of the capacitive bridge circuit allows the first comparator 342 and the second comparator 344 to determine the change in the value of at least one of the modelled conduit resistances and the transmitting electrode 314 and the receiving electrodes by determining differences between the reference voltage and the voltages at the first receiving terminal Rx.sub.1 and the second receiving terminal Rx.sub.2.

(61) It is noted that the second sensing electrode 318 and the second comparator 344 are optional, yet preferred for increasing accuracy of determining a flow through the beer conduit 312 and optionally the amount of the flow.

(62) In the description above for FIG. 3 and FIG. 4, it is noted that certain elements may be connected to a zero reference or a ground terminal. A ground terminal is defined as a terminal having a fixed voltage level—usually zero—relative to the earth. A zero reference level is a voltage level of one terminal of a voltage supply or another particular terminal in a circuit. In the implementations discussed above, this terminal is the lower terminal of voltage source V1 as depicted by FIG. 3. The zero reference is not always the same as the voltage level of the ground terminal (the ground level), as the zero reference may float relative to the ground level. The zero reference is only equal to the ground level if the zero reference terminal is connected to ground—like the protected earth of a wall socket or similar or equivalent.

(63) Referring to FIG. 3, the shielding electrodes are connected to the same level as the zero reference of V1, as is the terminal of the first reference capacitance C1 opposite to the reference terminal. These nodes may be connected to earth, but this does not have to be the case. The parasitic keg capacitance C7 is at the terminal opposite to the distal end of the beer conduit 312 to ground. The latter is the case, as the beer keg 306 is commonly placed in a beer cellar and/or refrigerator, away from the beer dispensing system 100, which does not provide a feasible option for connecting the zero reference to the environment of the beer keg 306.

(64) In another implementation, the signal provided on the first receiving electrode 316 and the second receiving electrode 318 is not compared to a voltage of the capacitive divider constituted by C1 and C2, but fed to the circuit depicted by FIG. 2A, with the signal on the receiving electrodes being provided to the trans impedance amplifier 206. Alternatively, an equivalent circuit may be provided, with or without the appropriate filters, in analogue or digital form. In such embodiment, the first reference capacitance C1 and the second reference capacitance C2 may be omitted. Additionally or alternatively, the first comparator 342 and the second comparator 344 may be omitted.

(65) Likewise, the electrodes depicted by FIG. 1 may also form part of a capacitive bridge circuit as depicted by FIG. 3 and FIG. 4.

(66) In summary, the aspects relate to a dispensing system for a beverage comprising in a tap system a bore for housing a duct. Along the bore, close to or on the duct, at least two electrodes are provided such that at least at some locations along the duct, the two electrodes are provided opposite to one another with the duct in between, thus constituting a capacitor. An oscillating signal, preferably having a sine waveform, is provided to one electrode and a signal is read out from the other electrode. As a beverage is drawn through the duct into a container, capacitance of the capacitor changes. The flowing beverage may have different characteristics, but capacitance may also change as the beverage in the duct is in conducting contact with a container that may be in contact with an earth contact. The change of capacitance results in a change of the amplitude of a detection circuit connected to the second electrode.

(67) In the description above, it will be understood that when an element such as layer, region or substrate is referred to as being “on” or “onto” another element, the element is either directly on the

other element, or intervening elements may also be present. Also, it will be understood that the values given in the description above, are given by way of example and that other values may be possible and/or may be strived for.

(68) Furthermore, the invention may also be embodied with less components than provided in the embodiments described here, wherein one component carries out multiple functions. Just as well may the invention be embodied using more elements than depicted in the Figures, wherein functions carried out by one component in the embodiment provided are distributed over multiple components.

(69) It is to be noted that the figures are only schematic representations of embodiments of the invention that are given by way of non-limiting examples. For the purpose of clarity and a concise description, features are described herein as part of the same or separate embodiments, however, it will be appreciated that the scope of the invention may include embodiments having combinations of all or some of the features described. The word ‘comprising’ does not exclude the presence of other features or steps than those listed in a claim. Furthermore, the words ‘a’ and ‘an’ shall not be construed as limited to ‘only one’, but instead are used to mean ‘at least one’, and do not exclude a plurality.

(70) A person skilled in the art will readily appreciate that various parameters and values thereof disclosed in the description may be modified and that various embodiments disclosed and/or claimed may be combined without departing from the scope of the invention.

(71) It is stipulated that the reference signs in the claims do not limit the scope of the claims, but are merely inserted to enhance the legibility of the claims.

Claims

1. A circuit for detecting a liquid flow through a duct of a tap for dispensing beverages, the circuit comprising: an alternating signal source having a signal terminal arranged to provide an alternating signal to a first electrode of a capacitive element comprising two electrodes provided adjacent to the duct along at least part of a length of the duct and the alternating signal source further comprising a zero terminal; a detection circuit arranged to be connected to the capacitive element for receiving an input signal from a second electrode of said two electrodes of the capacitive element and arranged to detect a signal amplitude value, the detection circuit being arranged to provide a detection signal based on the signal amplitude value at a detection terminal of the detection circuit; an electronic digital processing circuit arranged to: receive the detection signal; determine whether the detection signal satisfies a pre-determined criterion; and provide a flow signal if the pre-determined criterion is met; wherein: the detection circuit comprises a signal transformation circuit for transforming the input signal received from the second electrode, to provide a transformed signal having been transformed from a time domain to a frequency domain; the detection circuit further comprises a convolution module that provides an output that is a convolution of the transformed signal with an auxiliary signal; and the detection signal is based on the output of the convolution module, wherein the detection signal is indicative of said liquid flow flowing through the duct.

2. The circuit according to claim 1, wherein the alternating signal source is arranged to provide the alternating signal to the signal transformation circuit.

3. The circuit according to claim 2, wherein the alternating signal source provides a digital source signal and the circuit further comprises a digital to analogue transformer for converting the digital source signal to an analogue source signal and for providing the analogue source signal, as the alternating signal, to the first electrode.

4. The circuit according to claim 1, wherein the auxiliary signal is a step function.

5. The circuit according to claim 1, further comprising a temperature sensor for providing a temperature signal based on a sensed temperature, the temperature sensor being coupled to the

electronic digital processing circuit; wherein the pre-determined criterion is based on the temperature signal.

6. The circuit according to claim 1, further comprising an actuator connected to the electronic digital processing circuit, wherein the electronic digital processing circuit is arranged to actuate the actuator based on the flow signal.
7. A casing having an elongate bore having a proximal opening and a distal opening for housing and guiding at least part of a duct for a beverage dispensing tap, the casing comprising: the first electrode and the second electrode of the capacitive element of the circuit of claim 1, said first electrode and said second electrode being provided over at least respective first and second parts of a length of the bore.
8. The casing according to claim 7, wherein: the first electrode at least partially surrounds the bore; the second electrode at least partially surrounds the bore; and the first electrode and the second electrode are spaced apart along the length of the bore.
9. The casing according to claim 8 further comprising a third electrode between the first electrode and the second electrode, wherein the third electrode is arranged to be connected to a further detection circuit.
10. The casing according to claim 8, further comprising at least one shielding electrode between the first and second electrodes, the shielding electrode arranged to be connected to at least one of the zero terminal of the alternating signal source or to a ground terminal.
11. The casing according to claim 7, wherein the first electrode and the second electrode are provided substantially parallel to the bore.
12. A dispensing device for dispensing a beverage through a duct of a tap, the device comprising: the circuit according to claim 1; wherein the first electrode and the second electrode form the capacitive element coupled to the alternating signal source and to the detection circuit, the dispensing device further comprising: a casing having an elongate bore having a proximal opening and a distal opening for housing and guiding at least part of the duct, wherein the first electrode and the second electrode of the capacitive element are provided over respective first and second parts of a length of the bore.
13. The dispensing device according to claim 12, wherein the detection circuit further comprises a comparing terminal and is arranged to provide the detection signal based on a difference between a first signal value on the signal terminal and a second signal value on the comparing terminal, the dispensing device further comprising: a first capacitance provided between the comparing terminal and the signal terminal of the alternating signal source; and a second capacitance provided between the comparing terminal and at least one of the zero terminal or a ground terminal.
14. The dispensing device according to claim 12, wherein the processing circuit is arranged to: receive the detection signal; determine whether the detection signal satisfies a further pre-determined criterion; and provide a signal if the further pre-determined criterion is met for indicating that no duct is provided in the bore.
15. An apparatus comprising: a capacitive element coupled to a duct of a tap for dispensing beverages; an alternating signal circuit coupled to the capacitive element; a detection circuit configured to receive an input signal from the capacitive element, transform the input signal from a time domain signal to a frequency domain signal, convolve the frequency domain signal with an auxiliary signal to form a convolved signal, and output a detection signal responsive to information in the convolved signal, wherein the detection signal is indicative of a liquid flow flowing through the duct.
16. The apparatus of claim 15 wherein the capacitive element includes a first electrode and a second electrode, said first and second electrodes being at different positions along a length of the duct.
17. The apparatus of claim 15, wherein the alternating signal circuit is configured to provide an alternating signal to the first electrode of the capacitive element.

18. The apparatus of claim 17, wherein the detection circuit is configured to receive the input signal from the second electrode of the capacitive element.
 19. The apparatus of claim 15, wherein the detection signal is a transformation of the convolved signal in a time domain.
 20. The apparatus of claim 15, further comprising an electronic digital processing circuit configured for receiving the detection signal and, if the detection signal satisfies a pre-determined condition, providing a flow signal.
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